SP2022 CS-113 Lab 3 - Vending State Machine 2/27/2022

Write a C++ (OOP, Java, etc.) program that simulates the Vending State Machine. Your program will do the following:

Your program presents the user with a choice of your favorite beverages. Then allow the user to choose a beverage by entering a pair of input. The output which beverage user chooses.

Let I1 be a set of character inputs = {A, B, C, D, E, F, CLR, *}

Let I2 be a set of numeric inputs = {0, 1... 9}

Let A be a set of favorite beverages = {Coke, Pepsi, Water, 7UP}

Let B be a set of user input pair = {(A, 5), (B, 2), (D, 9), (F,7)}

Valid symbols:

$x \mathcal{E} I1$ *and* $y \mathcal{E} I2$

Valid selections:

$z \mathcal{E} A$ *and* $w \mathcal{E} B$

* Hint: Find valid user inputs

Input/output Table for the Vending State Machine						
		The truth table and The Boolean Expression Corresponding to a Circuit				
User Input from the keyboard	***Output***	***Input Signals***			***Output Signal***	***Output Display the beverage name and cost***
		C1	**C2**	**C3**	**O**	
	Invalid selection	0	0	0	0	Not available
(A,5)	Coke	0	0	1	1	$1.50
	Invalid selection	0	1	0	0	Not available
(B,2)	Pepsi	0	1	1	1	$1.75
	Invalid selection	1	0	0	0	Not available
(D,3)	Water	1	0	1	1	$1
(F,7)	7UP	1	1	0	1	$1.25
	Invalid selection	1	1	1	0	Not available

Your program should operate continuously to display the list of beverages and their costs, and user selections, like the picture below.

Example of the keypad:

Once a user has selected their drink the system should allow them to enter the amount of money they are inserting into the machine. The program should then calculate the amount of change to be returned and subtract one from the number of drinks in the machine. If the user selects a drink that has sold out an appropriate message should be displayed. Input validation must be performed out on the selection from the truth table. Array/vector must be used and an expected looping menu-driven of the keypad.

Once a user has selected their drink the system should allow users to enter the amount of money they are inserting into the machine. The program should then calculate the amount of change to be returned from the machine.